

Representing microgestures interaction for wearable computing

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ABSTRACT

Microgestures, i.e. fast and subtle movements of the fingers, offer always available and versatile interaction that makes them promising for wearable computing. While research has been done to define sets of microgestures and sensing systems, we do not know how to best represent microgestures and their associated commands which is essential for the adoption of microgesture interaction. Thus, this thesis focuses on how microgestures and their corresponding commands can be represented to users and discovered by them.

CCS CONCEPTS

• **Human-centered computing** → **Human computer interaction (HCI)**.

KEYWORDS

Microgestures; Command exploration; Visual cues; Representations.

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1 INTRODUCTION

If mid-air gestures quickly lead to a severe fatigue [16] and are not adapted to various contexts of use, microgestures are subtle, fast, easy to execute, versatile (eye-free, hand-free or not) and offer a better social acceptance than other gesture-based interactions [7, 8, 20]. To this day, many microgesture detection systems have been conceived but beyond the notion of availability, the question of democratisation arises. Sight being the most important human sense [10], we focus on visual systems.

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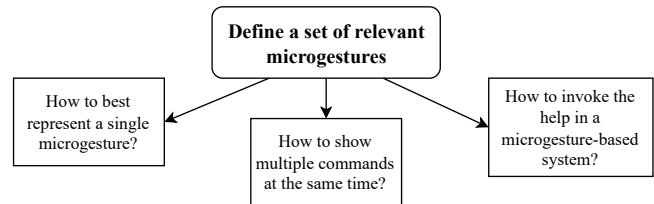


Figure 1: Research axis for the PhD.

How can a user figure out that microgestures are available to interact with a system and which ones? To what actions and commands are these microgestures associated to? How can a user trigger the appearance of a microgesture help system?

These are the three main research questions we address in this doctoral study. Facing the huge space of possibilities to explore, we made choices to delimit our research work.

2 POSITIONING OF OUR RESEARCH WORK

2.1 Microgestures in an nutshell

Microgestures are fast and subtle movements of the fingers for which Chaffangeon *et al.* [5] proposed μ Glyph, an expert notation to describe them with precision and consistency. Thus microgestures appear as a quick movement done by one or more fingers without moving the arm, sometimes including the wrist but often not. They are a private and subtle type of gesture [8] involving in majority thumb-based interactions [7] for which Soliman *et al.* [30] defined a design space.

A microgesture can be done while performing another task, whether with a free hand or while holding an object. As they can have a limited impact on the primary task [34], some studies pointed out their interest to be used while doing sport [4, 21] or with wearable technologies such as AR glasses [19, 20, 35], smartphones [37] and smartwatches [17].

In the light of existing studies [7, 20, 24, 29], **we chose to focus on the most commonly used microgestures: single-hand thumb-to-finger microgestures and finger flexions.**

2.2 Representation of gestures and their associated commands

Articles containing microgestures make use of a great diversity of representations but the relevance of these representations have not been studied yet. The taxonomies established by McAweeney *et al.* [26] and Antoine *et al.* [1] serve as a basis for our research. Moreover for studying any visual representation, the 7 variables identified by Bertin as part of the *Semiology of Graphics* [3] are relevant to consider. In addition, other studies pointed out the importance of animation [12, 25, 36] and distinguish *static*, i.e. not animated, and *dynamic*, i.e. animated, representations [2, 6, 11, 13].

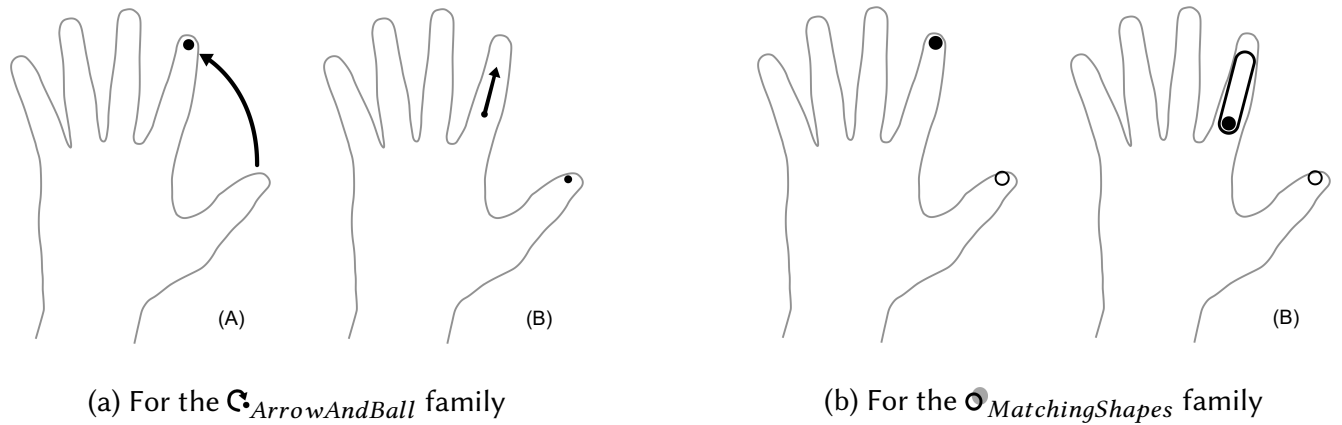


Figure 2: Representations of the (A) tap and (B) swipe up microgestures

As microgestures are envisioned to be used with 2D, e.g. smartwatches, and 3D, e.g. AR headsets, **we chose to study visual representations of microgestures usable in both contexts, e.g. on 2D drawings (i.e. cheatsheets for learning microgestures commands) or on 3D hands (i.e. augmented real hands or virtual hands).**

2.3 Studying the discoverability of a microgestures based interaction system

Discoverability can be defined as the ability for users to perceive and comprehend a system, function or input method as such when encountering it for the first time [23]. It can be about discovering the availability and/or the very nature of the interaction method [33] (*interaction discovery*) as well as about discovering a more or less hidden functionality [15, 27] (*feature discovery*). This is too broad of a topic to be fully covered, so **we chose to focus on the discovery of microgestures and their associated commands in a system that is known to allow interaction by microgestures.**

3 RESEARCH APPROACH

Having delimited our research work, many approaches could be envisioned. We chose to focus on the three axis presented in Figure 1.

3.1 Axis 1 : Representing a microgesture

μ Glyph is a visual notation defining a large microgesture vocabulary but it has been designed for experts. End-users need a *walk-up-and-use* system [33]. Thus arises the question of how to design a help system which represents both the microgesture to do and the support of this microgesture, i.e. the hand itself.

As a starting point, we have explored the visual cues currently used to represent microgestures. We identified a subset of 4 types of microgestures (tap, swipe, flex, hold) and have already designed and compared 21 groups of representations sharing consistent designs over our chosen microgestures. We chose to call those distinct groups "families" (see Figure 2a and 2b). This work allowed us to provide theoretical tools such as a metric and design guidelines centered on the representation of microgestures. **An article reporting two experimental studies on these representation families has been conditionally accepted at MobileHCI 2023.**

As our representations highly depend on the screen space, we need to further study the use of an help system in various contexts, e.g. smartwatches or AR glasses. To do so, we plan to focus on the posture of the hand and the degree of fidelity of the representations.

3.2 Axis 2 : Showing multiple commands at the same time

Once we know how to represent a microgesture, we still have to face two problems : how to represent multiple microgestures simultaneously and how to associate commands to their representations? Consequently, we have to establish the number of commands to show and the impact of the intrinsic proximity of their corresponding microgestures causing occlusion between their representations, e.g. tap and hold share the same initial touch phase.

We considered that showing the command is orthogonal to representing the associated microgesture. Consequently, we studied two cases : a command directly next to the representation or integrated in a legend.

Combined to the literature on visualization [9, 14, 18, 22, 31], we thus numbered 4 spatial and 4 non-spatial occlusion-aware design strategies. They are meant to improve the readability of the representation of multiple commands and reduce user's confusion. We are designing an experiment that would involve two sets of microgestures, namely tap-swipe and tap-hold, which would be distributed over one and then two fingers. The representations used in our experiment would come from two families to reduce potential biases : $\curvearrowright$ _{ArrowAndBall} (see Figure 2a) and $\odot$ _{MatchingShapes} (see Figure 2b). **This experiment is underway, and should yield its first results by the week of the conference.**

Chan *et al.* proposed a consensus set of 16 microgestures for 34 commands. This set does not include widely used microgestures such as the thumb tap on the middle phalanx of the index finger [17, 28, 32, 35]. Considering such an important set of possible commands, we also plan to conceive exploration strategies for help systems. Inspired by *swhidgets* [27], the hidden microgestures could be revealed by an exploration based on thumb flexes (cf Figure 3a) and fingers touching the other fingers (cf Figure 3b).

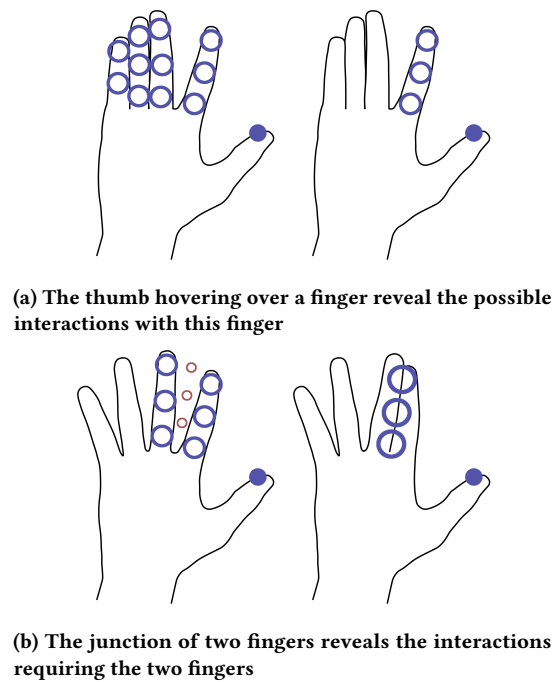


Figure 3: Interactions made visible by moving fingers

3.3 Axis 3 : Conceiving the invocation of the help system

In the related work of his book on public display interaction with body gestures [33], Walter distinguished 6 techniques for communicating gesture interactivity to the public. Associated with our representations and occlusion-aware design strategies (see preceding section), his work may shape various approaches on communicating the existence of a help system.

Our leading ideas consist in evaluating the user's reaction to multiple help invocation techniques with a serious game. This would allow us to explore many ideas and compare their performances in terms of user engagement, satisfaction, learning rate and recognition performances.

4 CONCLUSION

To sum up, this thesis is dedicated to how microgesture interaction can be learned and improved with either 2D or 3D representations. For the time being, 9 months of full-time work have been dedicated to the establishment of design guidelines to represent a single microgesture. The two remaining years will focus on the proposal of design and exploration strategies to represent multiple microgestures. We also propose to study how to invoke a help system designed for microgesture-based interactions.

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